MSG No. 115.288 $P_c\bar{4}m2$ [Type IV, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `-4m2`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,4,5,6,7,8]`
2	$\left[0, 0, \frac{1}{2}\right]$	`[9,10,11,12,13,14,15,16]`

Table 2: Wyckoff site: `2b`, site symmetry: `-4m2`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[1,2,3,4,5,6,7,8]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[9,10,11,12,13,14,15,16]`

Table 3: Wyckoff site: `2c`, site symmetry: `-4'm2'`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[1,2,7,8,11,12,13,14]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[3,4,5,6,9,10,15,16]`

Table 4: Wyckoff site: `2d`, site symmetry: `-4'm2'`

No.	position	mapping
1	$\left[0, 0, \frac{1}{4}\right]$	`[1,2,7,8,11,12,13,14]`
2	$\left[0, 0, \frac{3}{4}\right]$	`[3,4,5,6,9,10,15,16]`

Table 5: Wyckoff site: `4e`, site symmetry: `2mm.`

No.	position	mapping
1	$[0, 0, z]$	`[1,2,7,8]`
2	$[0, 0, -z]$	`[3,4,5,6]`
3	$\left[0, 0, z + \frac{1}{2}\right]$	`[9,10,15,16]`
4	$\left[0, 0, \frac{1}{2} - z\right]$	`[11,12,13,14]`

Table 6: Wyckoff site: `4f`, site symmetry: `2mm.`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[1,2,7,8]`
2	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[3,4,5,6]`
3	$\left[\frac{1}{2}, \frac{1}{2}, z + \frac{1}{2}\right]$	`[9,10,15,16]`
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2} - z\right]$	`[11,12,13,14]`

Table 7: Wyckoff site: `4g`, site symmetry: `2mm.`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,2,7,8]`
2	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[3,4,5,6]`
3	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,10,15,16]`
4	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	`[11,12,13,14]`

Table 8: Wyckoff site: `8h`, site symmetry: `..2`

No.	position	mapping
1	$[x,\ x,\ 0]$	`[1,3]`
2	$[-x,\ -x,\ 0]$	`[2,4]`
3	$[x,\ -x,\ 0]$	`[5,8]`
4	$[-x,\ x,\ 0]$	`[6,7]`
5	$\left[x,\ x,\ \frac{1}{2}\right]$	`[9,11]`
6	$\left[-x,\ -x,\ \frac{1}{2}\right]$	`[10,12]`
7	$\left[x,\ -x,\ \frac{1}{2}\right]$	`[13,16]`
8	$\left[-x,\ x,\ \frac{1}{2}\right]$	`[14,15]`

Table 9: Wyckoff site: `8i`, site symmetry: `..2'`

No.	position	mapping
1	$\left[x,\ x,\ \frac{1}{4}\right]$	`[1,11]`
2	$\left[-x,\ -x,\ \frac{1}{4}\right]$	`[2,12]`
3	$\left[x,\ x,\ \frac{3}{4}\right]$	`[3,9]`
4	$\left[-x,\ -x,\ \frac{3}{4}\right]$	`[4,10]`
5	$\left[x,\ -x,\ \frac{3}{4}\right]$	`[5,16]`
6	$\left[-x,\ x,\ \frac{3}{4}\right]$	`[6,15]`
7	$\left[-x,\ x,\ \frac{1}{4}\right]$	`[7,14]`
8	$\left[x,\ -x,\ \frac{1}{4}\right]$	`[8,13]`

Table 10: Wyckoff site: `8j`, site symmetry: `.m.`

No.	position	mapping
1	$[x,\ 0,\ z]$	`[1,8]`
2	$[-x,\ 0,\ z]$	`[2,7]`
3	$[0,\ x,\ -z]$	`[3,6]`
4	$[0,\ -x,\ -z]$	`[4,5]`
5	$\left[x,\ 0,\ z+\frac{1}{2}\right]$	`[9,16]`
6	$\left[-x,\ 0,\ z+\frac{1}{2}\right]$	`[10,15]`
7	$\left[0,\ x,\ \frac{1}{2}-z\right]$	`[11,14]`
8	$\left[0,\ -x,\ \frac{1}{2}-z\right]$	`[12,13]`

Table 11: Wyckoff site: 8k, site symmetry: .m.

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ z\right]$	[1,8]
2	$\left[-x,\ \frac{1}{2},\ z\right]$	[2,7]
3	$\left[\frac{1}{2},\ x,\ -z\right]$	[3,6]
4	$\left[\frac{1}{2},\ -x,\ -z\right]$	[4,5]
5	$\left[x,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[9,16]
6	$\left[-x,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[10,15]
7	$\left[\frac{1}{2},\ x,\ \frac{1}{2}-z\right]$	[11,14]
8	$\left[\frac{1}{2},\ -x,\ \frac{1}{2}-z\right]$	[12,13]

Table 12: Wyckoff site: 16l, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$[-x,\ -y,\ z]$	[2]
3	$[y,\ x,\ -z]$	[3]
4	$[-y,\ -x,\ -z]$	[4]
5	$[y,\ -x,\ -z]$	[5]
6	$[-y,\ x,\ -z]$	[6]
7	$[-x,\ y,\ z]$	[7]
8	$[x,\ -y,\ z]$	[8]
9	$\left[x,\ y,\ z+\frac{1}{2}\right]$	[9]
10	$\left[-x,\ -y,\ z+\frac{1}{2}\right]$	[10]
11	$\left[y,\ x,\ \frac{1}{2}-z\right]$	[11]
12	$\left[-y,\ -x,\ \frac{1}{2}-z\right]$	[12]
13	$\left[y,\ -x,\ \frac{1}{2}-z\right]$	[13]
14	$\left[-y,\ x,\ \frac{1}{2}-z\right]$	[14]
15	$\left[-x,\ y,\ z+\frac{1}{2}\right]$	[15]
16	$\left[x,\ -y,\ z+\frac{1}{2}\right]$	[16]